CONCEPT 54.2

Diversity and trophic structure characterize biological communities *(pp. 1222–1228)*

- **Species diversity** is affected by both the number of species in a community—its **species richness**—and their **relative abundance**.
- More diverse communities typically produce more **biomass** and show less year-to-year variation in growth than less diverse communities and are more resistant to introduced species.
- **Trophic structure** is a key factor in community dynamics. **Food chains** link the trophic levels from producers to top carnivores. Branching food chains and complex trophic interactions form **food webs**.
- **Foundation species** are large or abundant members of a community that provide food or habitat. **Keystone species** are usually less abundant species that exert a disproportionate influence on community structure. **Ecosystem engineers** influence community structure through their effects on the physical environment.
- In **bottom-up control**, the abundance of organisms at each trophic level is limited by nutrient supply or food availability. In **top-down control**, each trophic level is controlled by the abundance of consumers at higher trophic levels.

? *Based on indexes such as the Shannon diversity index, is a community of higher species richness always more diverse than a community of lower species richness? Explain.*

CONCEPT 54.3

Disturbance influences species diversity and composition *(pp. 1228–1231)*

- Increasing evidence suggests that **disturbance** and lack of equilibrium, rather than stability and equilibrium, are the norm for most communities. According to the **intermediate disturbance hypothesis**, moderate levels of disturbance can foster higher species diversity than can low or high levels of disturbance.
- **Ecological succession** is the sequence of community and ecosystem changes after a disturbance. **Primary succession** occurs in an area that is virtually lifeless, while **secondary succession** occurs after a disturbance has removed most but not all of the organisms in a community.

? *Is the disturbance pictured in Figure 54.28 more likely to initiate primary or secondary succession? Explain.*

CONCEPT 54.4

Biogeographic factors affect community diversity *(pp. 1231–1233)*

- Species richness generally declines along a latitudinal gradient from the tropics to the poles. Climate influences the diversity gradient through energy (heat and light) and water. The greater age of tropical environments also may contribute to their greater species richness.
- Species richness is directly related to a community's geographic size, a principle formalized in the **species-area curve**.
- Species richness on islands depends on island size and distance from the mainland. The island equilibrium model maintains that species richness on an ecological island reaches an equilibrium where new immigrations are balanced by extinctions.

? *How have periods of glaciation influenced latitudinal patterns of diversity?*

CONCEPT 54.5

Pathogens alter community structure locally and globally *(pp. 1234–1235)*

- **Pathogens** play a key in structuring terrestrial and marine communities.
- **Zoonotic pathogens** are transferred from other animals to humans and cause most emerging human diseases. Community ecology provides the framework for identifying key species interactions associated with such pathogens and for helping us track and control their spread.

? *Suppose a pathogen attacks a keystone species. Explain how this could alter the structure of a community.*

TEST YOUR UNDERSTANDING

For more multiple-choice questions, go to the **Practice Test** in the **Mastering Biology** eText or Study Area, or go to goo.gl/GruWRg.

Levels 1-2: Remembering/Understanding

1. The feeding relationships among the species in a community determine the community's
(A) secondary succession.
(B) ecological niche.
(C) species richness.
(D) trophic structure.

2. The principle of competitive exclusion states that
(A) two species cannot coexist in the same habitat.
(B) competition between two species always causes extinction or emigration of one species.
(C) two species that have exactly the same niche cannot coexist in a community.
(D) two species will stop reproducing until one species leaves the habitat.

3. Based on the intermediate disturbance hypothesis, a community's species diversity is increased by
(A) frequent massive disturbance.
(B) stable conditions with no disturbance.
(C) moderate levels of disturbance.
(D) human intervention to eliminate disturbance.

4. According to the island equilibrium model, species richness would be greatest on an island that is
(A) large and remote.
(B) small and remote.
(C) large and close to a mainland.
(D) small and close to a mainland.

Levels 3-4: Applying/Analyzing

5. Predators that are keystone species can maintain species diversity in a community if they
(A) competitively exclude other predators.
(B) prey on the community's dominant competitors.
(C) reduce the number of disruptions in the community.
(D) prey on the least abundant species in the community.

6. Food chains tend to be short because
(A) only a single species of herbivore feeds on each plant species.
(B) local extinction of a species causes extinction of the other species in its food chain.
(C) most of the energy in a trophic level is lost as energy passes to the next higher level.
(D) most producers are inedible.